Taewon Kim
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Education

- **Graduate School of AI, KAIST** — Seoul, Korea
  PhD candidate advised by Sungsoo Ahn; GPA: 4.21/4.3 — *Mar. 2024 – Present*

- **Korea Advanced Institute of Science and Technology** — Daejeon, Korea
  Major in Computer Science, Electrical Engineering; GPA: 3.78/4.3 — *Mar. 2018 – Mar. 2024*

Selected Publications

- **[NeurIPS 26] Atom-level Protein Representation Learning Improves Protein Structure Prediction**
  ***Taewon Kim**, Hyosoon Jang, Hyunjin Seo, Seonghwan Seo, Hyeongwoo Kim, Wonho Zhung, Mingyeong Shin, Wooyoun Kim, Sungsoo Ahn*
  First to show that apo-structure representations enhance co-folding performance, and proposed TriProRep, a scalable framework for protein structure representation learning that directly benefits co-folding models.

- **[NeurIPS 26] A Systematic Evaluation of Co-folding Model Representations for Small-Molecule Learning**
  *Hyosoon Jang, Hyunjin Seo, Honghui Kim, Seonghyun Park, **Taewon Kim**, Yunhui Jang, Sungsoo Ahn*
  Showed that atom-level ligand representations from co-folding models rival or beat dedicated models on ADMET benchmarks, while accelerating molecular generation and structure-guided ligand optimization.

- **[ICML 26, ICLRW Oral 26] DNAChunker: Learnable Tokenization for DNA Language Models**
  ***Taewon Kim**, Jihwan Shin, Hyomin Kim, Youngmok Jung, Jonghoon Lee, Won-Chul Lee, Sungsoo Ahn, Insu Han*
  Motivated by the lack of study in tokenization methods specific for DNA sequences which lack natural separators such as spaces, proposed a learnable tokenization method for DNA sequence.

- **[ICLR 26] Learning Flexible Forward Trajectories for Masked Molecular Diffusion**
  Hyunjin Seo, **Taewon Kim***, Sungsoo Ahn*
  Introduced a flexible noise scheduling framework for masked diffusion models, significantly enhancing the validity of masked diffusion models in molecules, which have previously shown to have diminishing results.

- **[ICLR 25] REBIND: Enhancing Ground-state Molecular Conformation Prediction via Force-Based Graph Rewiring**
  Taewon Kim, Hyunjin Seo*, Sungsoo Ahn, Eunho Yang*
  Developed a graph rewiring technique incorporating physical force into attention bias for accurate ground-state molecular conformation prediction.

Preprints

- **Co-folding with a Soup of Representations**
  *Hyosoon Jang, **Taewon Kim**, Sungsoo Ahn*
  Introduced SoupFold, which transfers complementary representations among independently trained co-folding models to improve protein-protein and protein-ligand structure prediction without retraining.

Publications in General Domain

- **[WACV 26] Towards Reliable Test-Time Adaptation: Style Invariance as a Correctness Likelihood**
  Gilhyun Nam, **Taewon Kim***, Eunho Yang*

- **[Preprint] Unleashing the Potential of Text-Attributed Graphs: Automatic Relation Decomposition via Large Language Models**
  *Hyunjin Seo, **Taewon Kim**, June Yong Yang, Eunho Yang*

- **[NeurIPSW 24] AdapTable: Test-Time Adaptation for Tabular Data via Shift-Aware Uncertainty Calibrator**
  Changhun Kim, **Taewon Kim***, Seungyeon Woo, June Yong Yang, Eunho Yang*

- **[NeurIPS 22] NOTE: Robust Continual Test-time Adaptation Against Temporal Correlation**
  *Taesik Gong, Jongheon Jeong, **Taewon Kim**, Yewon Kim, Jinwoo Shin, Sung-Ju Lee*

Invited Talks

- **LG Materials Intelligence Lab** — Jun. 2025
  Learning Flexible Forward Trajectories for Masked Molecular Diffusion — *Seoul, South Korea*
- **KAIST-MILA Prefrontal AI Center** — Jun. 2026
  K-Fold and Atom-level Protein Representation Learning
- **Multiomics Reading Group** — Jul. 2026
  Atom-level Protein Representation Learning Improves Protein Structure Prediction

Research Experience

- **MLI Lab, KAIST** — Daejeon, Korea
  Research Intern — *Jun. 2023 – Jan. 2024*
- **MIIL Lab, KAIST** — Daejeon, Korea
  Research Intern — *Jun. 2022 – Jun. 2023*
- **INA Lab, KAIST** — Daejeon, Korea
  Research Intern — *Feb. 2020 – Jun. 2020*

Projects

- **AI-Specialized Foundation Model Development Project (Oct. 2025 - Present)**: Participating in project to surpass AlphaFold, developing a general model capable of predicting both apo and holo structures of proteins. Within it, currently developing a I-JEPA inspired protein representation learning framework
- **Genetic Language Model for Cancer Subtype Classification (Jul. 2025 - Present)**: Based upon the DNA language model developed via learnable tokenization method, incorporating it with real-world patient genomics data to create an AI framework to classify patients' cancer subtypes.

Awards

- **22 QualComm Innovative Fellowship Korea Finalist**: Oct. 2022
- **Deans' List for Electrical Engineering**: Spring 2022
- **K-Startup Challenge Finalist**: Fall 2020

Academic Service

- **Conference Reviewer**: ICLR 2026, ICML 2026, NeurIPS 2026
  Workshop Reviewer: ICML 2026 Workshop on AI for Science (AI4Science)

Skills

- **Programming**: Python, Pytorch, Tensorflow, Tensorflowlite, Android Studio, React
- **Language**: English (Served as translator in Korea Army), Chinese